2017 AMC 12B Problem 20: Strategic Analysis

Using AMC12/COMC Problem-Solving Framework

Problem Statement

Real numbers x and y are chosen independently and uniformly at random from the interval $(0, 1)$. What is the probability that $\lfloor \log_2 x \rfloor = \lfloor \log_2 y \rfloor$?

Answer Choices:

(A) $\frac{1}{8}$ **(B)** $\frac{1}{6}$ **(C)** $\frac{1}{4}$ **(D)** $\frac{1}{3}$ **(E)** $\frac{1}{2}$

PROBLEM CONFIGURATION ANALYSIS

A. Problem Classification

- **Problem Type**: To Find – determine a probability value
- **Difficulty Band**: Late-band (Problem 20 of 25) – requires understanding of logarithms, floor functions, and geometric series
- **Competition Context**: AMC12 (75 min, 25 problems) – approximately 3-4 minutes allocated for late-band problems
- **Mathematical Domain**: Probability with strong connections to logarithms, floor functions, and infinite series

B. Problem Structure Identification

- **Given Information**:
  - x, y chosen independently and uniformly from $(0, 1)$
  - Need probability that $\lfloor \log_2 x \rfloor = \lfloor \log_2 y \rfloor$
- **Constraints**:
  - Open interval $(0, 1)$ means $0 < x, y < 1$
  - For $0 < x < 1$: $\log_2 x < 0$ (always negative)
  - Floor function creates discrete integer values from continuous logarithms
  - Independence: joint probability is product of marginal probabilities
- **Objective**: Find $P(\lfloor \log_2 x \rfloor = \lfloor \log_2 y \rfloor)$
- **Hidden Assumptions**:
  - The interval $(0, 1)$ partitions naturally based on floor values
  - Each floor value corresponds to a dyadic interval: $[\frac{1}{2^{n+1}}, \frac{1}{2^n})$

- These intervals have lengths that form a geometric sequence
- The probability involves an infinite series

- **Answer Choice Leverage**:
  - All choices are unit fractions with small denominators
  - $\frac{1}{2}$ would be too high (requires strong correlation)
  - $\frac{1}{8}$ seems too low (there's overlap)
  - $\frac{1}{3}$ and $\frac{1}{4}$ are geometric series sums ($\frac{1}{1-r}$ pattern)
  - Can eliminate extreme values and test middle options

C. Strategic Orientation

- **Hypothesis**: Let $X, Y \sim \text{Uniform}(0, 1)$ be independent
- **Conclusion**: Determine $P(\lfloor \log_2 X \rfloor = \lfloor \log_2 Y \rfloor)$
- **Notation**:
  - X, Y: random variables uniformly distributed on $(0, 1)$
  - $\lfloor \cdot \rfloor$: floor function (greatest integer function)
  - $\log_2$: logarithm base 2
  - For $x \in (0, 1)$: $\lfloor \log_2 x \rfloor \in \{-1, -2, -3, \ldots\}$
- **Intuitive Approach**:
  - Partition $(0, 1)$ by floor values of $\log_2$
  - $\lfloor \log_2 x \rfloor = -1$ when $x \in [\frac{1}{2}, 1)$ (length $\frac{1}{2}$)
  - $\lfloor \log_2 x \rfloor = -2$ when $x \in [\frac{1}{4}, \frac{1}{2})$ (length $\frac{1}{4}$)
  - Pattern: each interval has length $\frac{1}{2^n}$
  - Sum probabilities that both fall in same interval
- **Cross-Domain Potential**:
  - **Geometric**: Visualize as areas in coordinate plane
  - **Algebraic**: Use series summation formulas
  - **Analysis**: Understand interval structure through logarithm properties

STRATEGIC FRAMEWORK APPLICATION

A. Psychological Strategies

- **Mental Toughness**: Don't be intimidated by the nested functions (floor of logarithm); break down step-by-step
- **Creativity Cultivation**: Recognize that floor function creates discrete regions from continuous domain; visualize interval partitioning

- **Iterative Refinement**: Start with small cases ($n = -1, -2$), identify pattern, then generalize to infinite series

B. Investigation Strategies

- **Startup Orientation**: Recognize compound function structure: probability $\to$ floor function $\to$ logarithm
- **Get Your Hands Dirty**: Test specific values to understand floor behavior
  - $x = 0.75 \in [\frac{1}{2}, 1)$: $\log_2(0.75) \approx -0.415$, so $\lfloor \log_2(0.75) \rfloor = -1$
  - $x = 0.3 \in [\frac{1}{4}, \frac{1}{2})$: $\log_2(0.3) \approx -1.737$, so $\lfloor \log_2(0.3) \rfloor = -2$
- **Penultimate Step**: Once we identify the geometric series pattern, the final step is applying the formula $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$
- **Wishful Thinking**: We wish for a simple pattern in interval lengths – and indeed, powers of $\frac{1}{2}$ emerge!
- **Make It Easier**: Instead of thinking about logarithms, think about powers of 2: $2^{-1} = \frac{1}{2}$, $2^{-2} = \frac{1}{4}$, etc.
- **Conjecture via Small Cases**:
  - Case $n = -1$: $P(\text{both in } [\frac{1}{2}, 1)) = (\frac{1}{2})^2 = \frac{1}{4}$
  - Case $n = -2$: $P(\text{both in } [\frac{1}{4}, \frac{1}{2})) = (\frac{1}{4})^2 = \frac{1}{16}$
  - Pattern: probabilities are $\frac{1}{4}, \frac{1}{16}, \frac{1}{64}, \ldots$ (geometric with ratio $\frac{1}{4}$)
- **Visualization**: Draw number line $(0, 1)$ partitioned at $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots$; or draw unit square $(0, 1) \times (0, 1)$ with diagonal stripes for matching floor values

C. Late-Band Strategic Playbook

- **Answer-Choice Leverage**: Recognize $\frac{1}{3}$ suggests geometric series with $r = \frac{1}{4}$ (since $\frac{a}{1-r} = \frac{1/4}{1-1/4} = \frac{1/4}{3/4} = \frac{1}{3}$)
- **Make It Easier**: Convert logarithm problem to interval problem; convert continuous to discrete (via floor)
- **Penultimate Step**: Identify that answer is $\sum_{n=1}^{\infty} (\frac{1}{2^n})^2 = \sum_{n=1}^{\infty} \frac{1}{4^n}$
- **Wishful Thinking**: Hope for geometric series – and it appears!
- **Conjecture via Small Cases**: Compute first few terms, recognize pattern, apply formula

D. Argument Construction Strategy

- **Direct Proof**:
  1. Partition $(0, 1)$ by floor values
  2. Compute probability for each case
  3. Sum using geometric series formula
- **Proof by Contradiction**: Not applicable
- **Mathematical Induction**: Could verify pattern, but direct summation is cleaner
- **Stylistic Conventions**: Use case analysis; leverage independence and uniform distribution

E. Cross-Domain Translation

- **Draw a Picture**: Unit square with diagonal stripes (regions where floor values match)
- **Recast the Problem**: "Interval matching" problem rather than "floor of logarithm" problem
- **Change Your Point of View**: Think in terms of powers of 2 rather than logarithms

TACTICAL METHODOLOGY SELECTION

A. Core Mathematical Tactics

- **Symmetry**: Independence and identical distributions create symmetry
- **Extreme Principle**: Not directly applicable
- **Invariants**: Floor value partitions are invariant under the problem structure
- **Monovariants**: Not applicable
- **Parity**: Not applicable
- **Graph Theory**: Not applicable
- **Generating Functions**: Not primary, though series summation is related
- **Pattern Recognition**: Identify geometric sequence in interval lengths and probabilities

B. Crossover Techniques

- **Algebraic-Probabilistic**: Convert probability question to series summation
- **Geometric-Probabilistic**: Use geometric interpretation via unit square regions
- **Analytic-Discrete**: Floor function converts continuous logarithm to discrete integer values

C. Domain-Specific Approaches

- **Probability**: Independence, uniform distribution, total probability (sum over disjoint cases)
- **Analysis**: Logarithm properties, floor function behavior, interval decomposition
- **Algebra**: Geometric series summation
- **Geometry**: Unit square representation (optional but helpful for visualization)

CONVERSION TECHNIQUES APPLICATION

A. Technique Recognition Index

- **T17-GeoProb**: Geometric Probability – applicable for unit square visualization
- **Series Summation**: Geometric series formula – PRIMARY technique
- **Case Analysis**: Partition by floor values – essential structure
- **Logarithm Properties**: Understanding $\log_2 x$ for $x \in (0, 1)$

B. Technique Selection

Primary Technique: Case Analysis + Geometric Series
Why it applies:

- Floor function naturally partitions $(0, 1)$ into countably many intervals
- Each interval corresponds to one floor value
- Interval lengths form geometric sequence: $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots$
- Probabilities form geometric sequence: $\frac{1}{4}, \frac{1}{16}, \frac{1}{64}, \ldots$

Configuration cues: Floor of logarithm with uniform random variables
Objective cues: Finding probability of equality
Supporting Technique: Geometric Series Formula
For $|r| < 1$: $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$
In our case: $a = \frac{1}{4}$, $r = \frac{1}{4}$, giving $\frac{1/4}{1-1/4} = \frac{1/4}{3/4} = \frac{1}{3}$

C. Technique Integration

1. **Understand domain**: For $x \in (0, 1)$, $\log_2 x$ ranges from $-\infty$ to 0
2. **Identify intervals**: $\lfloor \log_2 x \rfloor = n$ when $x \in [2^n, 2^{n+1})$ for $n \in \{\ldots, -3, -2, -1\}$
3. **For our interval**:
   - $\lfloor \log_2 x \rfloor = -1$ when $x \in [\frac{1}{2}, 1)$ (length $\frac{1}{2}$)
   - $\lfloor \log_2 x \rfloor = -2$ when $x \in [\frac{1}{4}, \frac{1}{2})$ (length $\frac{1}{4}$)

- $\lfloor \log_2 x \rfloor = -k$ when $x \in [\frac{1}{2^k}, \frac{1}{2^{k-1}})$ (length $\frac{1}{2^k}$)

4. **Compute probability**: For both x and y in same interval with length ℓ: $P = \ell \cdot \ell = \ell^2$
5. **Sum over all cases**: $\sum_{k=1}^{\infty} (\frac{1}{2^k})^2 = \sum_{k=1}^{\infty} \frac{1}{4^k} = \frac{1/4}{1-1/4} = \frac{1}{3}$

D. Conflict Resolution

No conflicts arise. The approach is:

- Cases are mutually exclusive (disjoint intervals)
- Cases are exhaustive (cover all of $(0, 1)$)
- Probabilities properly computed via independence
- Series converges (geometric with $|r| = \frac{1}{4} < 1$)

KEY INSIGHT

The Deep Insight:
The key insight is recognizing that the floor function partitions $(0, 1)$ into dyadic intervals (powers of 1/2): $[\frac{1}{2}, 1), [\frac{1}{4}, \frac{1}{2}), [\frac{1}{8}, \frac{1}{4}), \ldots$
Each interval has length $\frac{1}{2^k}$ for $k = 1, 2, 3, \ldots$
The probability that both x and y fall in the same interval is: $P = \sum_{k=1}^{\infty} \left(\frac{1}{2^k}\right)^2 = \sum_{k=1}^{\infty} \frac{1}{4^k} = \frac{1/4}{1-1/4} = \frac{1}{3}$
This is a geometric series with first term $\frac{1}{4}$ and common ratio $\frac{1}{4}$.
Why This is Elegant:
This problem beautifully connects three mathematical concepts:

- **Logarithms**: Map multiplicative structure (powers of 2) to additive structure (integers)
- **Floor function**: Discretizes continuous range into countable cases
- **Geometric series**: The dyadic structure naturally produces a geometric sequence

The base-2 logarithm is particularly natural here because it reveals the binary structure of $(0, 1)$, where each "bit position" corresponds to one dyadic interval.

VERIFICATION STRATEGY DESIGN

A. Verification Level Selection

This is Problem 20 (late-band), so use **Level 3: Comprehensive Verification**
Decision criteria:

- High difficulty (P20/25)
- Multiple conceptual steps (logarithms, floor, probability, series)

- Requires correct understanding of interval structure
- Series summation must be accurate

B. Specialized Verification Methods

- **Interval Check**:
  - Verify $\lfloor \log_2(\frac{1}{2}) \rfloor = \lfloor -1 \rfloor = -1$ ✓
  - Verify $\lfloor \log_2(\frac{1}{4}) \rfloor = \lfloor -2 \rfloor = -2$ ✓
  - Verify intervals partition $(0, 1)$: $\sum_{k=1}^{\infty} \frac{1}{2^k} = 1$ ✓
- **Series Computation**:
  - First term: $(\frac{1}{2})^2 = \frac{1}{4}$ ✓
  - Common ratio: $\frac{1}{16} \div \frac{1}{4} = \frac{1}{4}$ ✓
  - Sum: $\frac{1/4}{1-1/4} = \frac{1/4}{3/4} = \frac{1}{3}$ ✓
- **Boundary Cases**:
  - What if both in $[\frac{1}{2}, 1)$? Probability $= \frac{1}{4}$ (first term) ✓
  - As $k \to \infty$, terms $\to 0$ (series converges) ✓
- **Answer Choice Verification**:
  - Check answer $\frac{1}{3}$: This equals geometric series with $a = \frac{1}{4}$, $r = \frac{1}{4}$ ✓
  - Sanity check: $0 < \frac{1}{3} < \frac{1}{2}$ (probability is positive and less than $\frac{1}{2}$) ✓
- **Alternative Verification via Geometric Probability**:
  - Draw unit square $(0, 1) \times (0, 1)$
  - Favorable region: union of squares $[\frac{1}{2^{k+1}}, \frac{1}{2^k}) \times [\frac{1}{2^{k+1}}, \frac{1}{2^k})$
  - Total area: $\sum_{k=1}^{\infty} (\frac{1}{2^k})^2 = \frac{1}{3}$ ✓

C. Confidence Calibration

Confidence Level: 95% (High)
Reasoning:

- Interval structure is correct (verified with specific values)
- Series computation is standard and reliable
- Answer matches expected form (unit fraction from geometric series)
- Multiple verification methods agree

Potential concerns: None significant; approach is sound and well-verified

WARNING: Common Pitfall

Common Pitfall #1: Forgetting that $\log_2 x < 0$ for $x \in (0, 1)$. Students may incorrectly think floor values are positive.
Common Pitfall #2: Misidentifying interval boundaries. Remember:

- $\lfloor \log_2 x \rfloor = -1$ for $x \in [\frac{1}{2}, 1)$, NOT $[\frac{1}{2}, 1]$
- Interval is half-open: $[2^{-k}, 2^{-k+1})$

Common Pitfall #3: Incorrect geometric series formula. Be careful with indexing:

- $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$ (starting from $n = 0$)
- $\sum_{n=1}^{\infty} ar^n = \frac{ar}{1-r}$ (starting from $n = 1$)

In our problem: $\sum_{k=1}^{\infty} \frac{1}{4^k}$ can be computed as:

- Method 1: $\sum_{k=1}^{\infty} 1 \cdot (\frac{1}{4})^k = \frac{1 \cdot \frac{1}{4}}{1-\frac{1}{4}} = \frac{1/4}{3/4} = \frac{1}{3}$ ✓
- Method 2: $\sum_{k=0}^{\infty} (\frac{1}{4})^k - (\frac{1}{4})^0 = \frac{1}{1-\frac{1}{4}} - 1 = \frac{4}{3} - 1 = \frac{1}{3}$ ✓

Common Pitfall #4: Thinking the problem asks for $P(x = y)$ (probability of exact equality). The floor function makes this a discrete matching problem, not a continuous one.

PRO TIP

Pro Tip #1: When you see floor of logarithm, immediately think about interval partitioning by powers of the base.
Pro Tip #2: For $x \in (0, 1)$, $\log_b x$ is always negative. The floor values are $-1, -2, -3, \ldots$
Pro Tip #3: Dyadic intervals (powers of 2) are natural for base-2 logarithms. This creates geometric sequences in both interval lengths and probabilities.
Pro Tip #4: Geometric series with ratio $r = \frac{1}{4}$ gives sum $\frac{a}{3/4} = \frac{4a}{3}$. If first term is $\frac{1}{4}$, sum is $\frac{1}{3}$.
Pro Tip #5: Visualize this problem as a unit square $[0, 1] \times [0, 1]$ in the (x, y) plane. The favorable region consists of small squares along the "diagonal" where floor values match:

- Square 1: $[\frac{1}{2}, 1) \times [\frac{1}{2}, 1)$ with area $\frac{1}{4}$
- Square 2: $[\frac{1}{4}, \frac{1}{2}) \times [\frac{1}{4}, \frac{1}{2})$ with area $\frac{1}{16}$
- Square 3: $[\frac{1}{8}, \frac{1}{4}) \times [\frac{1}{8}, \frac{1}{4})$ with area $\frac{1}{64}$
- And so on...

Total favorable area $= \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \cdots = \frac{1}{3}$

SOLUTION ROADMAP

Solution Roadmap

Step 1: Understand the Floor Function Behavior

- For $x \in (0,1)$: $\log_2 x < 0$ (negative)
- $\lfloor \log_2 x \rfloor$ takes values $\ldots, -3, -2, -1$
- $\lfloor \log_2 x \rfloor = n \iff n \le \log_2 x < n+1 \iff 2^n \le x < 2^{n+1}$

Step 2: Partition $(0,1)$ by Floor Values

Key Understanding: $\lfloor \log_2 x \rfloor = n$ means $n \le \log_2 x < n+1$, which is equivalent to $2^n \le x < 2^{n+1}$

- $\lfloor \log_2 x \rfloor = -1$ when $x \in [\frac{1}{2}, 1)$
  - Derivation: $-1 \le \log_2 x < 0 \iff 2^{-1} \le x < 2^0 \iff \frac{1}{2} \le x < 1$
  - Check: $\log_2(1/2) = -1$, $\log_2(1) = 0$ ✓
  - Interval length: $1 - \frac{1}{2} = \frac{1}{2}$
  - Probability: $P(x \in [\frac{1}{2}, 1)) = \frac{1}{2}$
- $\lfloor \log_2 x \rfloor = -2$ when $x \in [\frac{1}{4}, \frac{1}{2})$
  - Derivation: $-2 \le \log_2 x < -1 \iff 2^{-2} \le x < 2^{-1} \iff \frac{1}{4} \le x < \frac{1}{2}$
  - Check: $\log_2(1/4) = -2$, $\log_2(1/2) = -1$ ✓
  - Interval length: $\frac{1}{2} - \frac{1}{4} = \frac{1}{4}$
  - Probability: $P(x \in [\frac{1}{4}, \frac{1}{2})) = \frac{1}{4}$
- **General Pattern**: $\lfloor \log_2 x \rfloor = -k$ when $x \in [\frac{1}{2^k}, \frac{1}{2^{k-1}})$
  - Interval length: $\frac{1}{2^{k-1}} - \frac{1}{2^k} = \frac{2-1}{2^k} = \frac{1}{2^k}$
  - Probability: $P(x \in [\frac{1}{2^k}, \frac{1}{2^{k-1}})) = \frac{1}{2^k}$
- **Coverage Check**: These intervals partition $(0,1)$ completely:

$$\sum_{k=1}^{\infty} \frac{1}{2^k} = \frac{1/2}{1-1/2} = \frac{1/2}{1/2} = 1 ✓$$

Step 3: Compute Probability for Each Case

- Since x and y are independent and uniform on $(0,1)$:
- $P(\lfloor \log_2 x \rfloor = \lfloor \log_2 y \rfloor = -k)$
- $= P(x \in [\frac{1}{2^k}, \frac{1}{2^{k-1}})) \cdot P(y \in [\frac{1}{2^k}, \frac{1}{2^{k-1}}))$
- $= \frac{1}{2^k} \cdot \frac{1}{2^k} = \frac{1}{4^k}$

Specific Cases:

- $k = 1$: $P(\text{both in } [\frac{1}{2}, 1)) = (\frac{1}{2})^2 = \frac{1}{4}$
- $k = 2$: $P(\text{both in } [\frac{1}{4}, \frac{1}{2})) = (\frac{1}{4})^2 = \frac{1}{16}$
- $k = 3$: $P(\text{both in } [\frac{1}{8}, \frac{1}{4})) = (\frac{1}{8})^2 = \frac{1}{64}$

Step 4: Sum Over All Cases

- By law of total probability (cases are disjoint):

$$\begin{aligned}
P(\lfloor \log_2 x \rfloor = \lfloor \log_2 y \rfloor) &= \sum_{k=1}^{\infty} P(\text{both have floor value } -k) \\
&= \sum_{k=1}^{\infty} \frac{1}{4^k} \\
&= \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \cdots \\
&= \frac{1}{4}(1 + \frac{1}{4} + \frac{1}{16} + \cdots) \\
&= \frac{1}{4} \cdot \frac{1}{1 - \frac{1}{4}} \\
&= \frac{1}{4} \cdot \frac{4}{3} \\
&= \frac{1}{3}
\end{aligned}$$

Step 5: Verify

- Check: Geometric series with $a = \frac{1}{4}$, $r = \frac{1}{4}$
- Sum: $\frac{a}{1-r} = \frac{1/4}{3/4} = \frac{1}{3}$ ✓
- Answer: $\boxed{\textbf{(D)}\ \frac{1}{3}}$

Alternative Approach: Geometric Probability

Visualization Method:

- Draw unit square $[0, 1] \times [0, 1]$ representing (x, y) pairs
- Favorable region: union of squares where floor values match
- Each square: $[\frac{1}{2^{k+1}}, \frac{1}{2^k}) \times [\frac{1}{2^{k+1}}, \frac{1}{2^k})$ with area $\frac{1}{4^k}$
- Total favorable area: $\sum_{k=1}^{\infty} \frac{1}{4^k} = \frac{1}{3}$
- Probability: $\frac{\text{favorable area}}{\text{total area}} = \frac{1/3}{1} = \frac{1}{3}$

Comprehensive Pitfall Guide

1. **Sign Error**: Remember $\log_2 x < 0$ for $x \in (0, 1)$. The floor values are negative: $-1, -2, -3, \ldots$
2. **Interval Boundaries**: Use $[2^{-k}, 2^{-k+1})$ (left-closed, right-open), not $(2^{-k}, 2^{-k+1}]$. The floor function makes the left endpoint inclusive.
3. **Series Indexing**: Our series starts from $k = 1$, not $k = 0$, because the first interval is $[\frac{1}{2}, 1)$ (corresponding to $k = 1$)
4. **Independence**: Since x and y are independent, we must square probabilities: $P(\text{both in interval}) = P(x \text{ in interval}) \times P(y \text{ in interval}) = \ell^2$ where ℓ is the interval length
5. **Geometric Series Formula**: For $\sum_{k=1}^{\infty} \frac{1}{4^k}$, apply the formula for series starting at $k = 1$: $\sum_{k=1}^{\infty} \frac{1}{4^k} = \frac{1/4}{1-1/4} = \frac{1}{3}$ Don't use $\frac{1}{1-1/4} = \frac{4}{3}$ (that's for starting at $k = 0$)
6. **Interval Length Calculation**: The length of $[\frac{1}{2^k}, \frac{1}{2^{k-1}})$ is $\frac{1}{2^{k-1}} - \frac{1}{2^k} = \frac{1}{2^k}$, not $\frac{1}{2^{k-1}}$

COMPETITION STRATEGY INTEGRATION

A. Time Management

- **Estimated Time**: 3-4 minutes for late-band problem
- **Breakdown**:
  - Understanding floor and logarithm behavior: 45 seconds
  - Identifying interval structure: 60 seconds
  - Computing first few probabilities: 45 seconds
  - Recognizing geometric series and summing: 45 seconds
  - Verification: 15 seconds
- **If stuck**: Move to next problem after 4 minutes; return if time remains

B. Problem Prioritization

- This is a probability problem with logarithms – prioritize if you're strong in both areas
- Requires comfort with floor functions and geometric series
- If you don't immediately see the interval partition, this problem can be time-consuming
- Skip if you're uncomfortable with logarithms or series; return later

C. Risk Assessment

- **Risk Level**: Medium-High
- **Confidence needed**: Must understand floor function behavior and geometric series

- **Abandonment criterion**: If you don't see interval partition pattern within 90 seconds, move on

D. Strategic Notes

- This problem rewards pattern recognition (dyadic intervals, geometric series)
- The compound function (floor of logarithm) is deliberately complex – don't panic
- Testing small cases ($k = 1, 2, 3$) quickly reveals the pattern
- Once you identify geometric series with $r = \frac{1}{4}$, the answer is immediate

Summary

This problem exemplifies late-band AMC12 probability: it combines multiple concepts (uniform distributions, logarithms, floor functions, infinite series) in a non-obvious way. The key insights are:

1. The floor function partitions $(0, 1)$ into dyadic intervals
2. Each interval has length $\frac{1}{2^k}$ for $k = 1, 2, 3, \ldots$
3. Probabilities form a geometric series: $\frac{1}{4}, \frac{1}{16}, \frac{1}{64}, \ldots$
4. The sum is $\frac{1/4}{1-1/4} = \frac{1}{3}$

With practice recognizing floor-of-logarithm patterns and geometric series, problems like this become accessible in 3-4 minutes.